

1.12 Numerical Methods

Mathematics B (MEI) — OCR B — A-Level (H640) — Pure Mathematics — spec refs e1–e5, c34–c35

What this topic covers. Locating roots by change of sign and the circumstances in which that fails, fixed point iteration with staircase and cobweb diagrams, the Newton-Raphson method and its failures, and the trapezium rule with a justified over- or under-estimate.

1. Why numerical methods

Most equations have no exact algebraic solution. Numerical methods produce an **approximation** to a stated accuracy, and part of the skill is knowing when they can be trusted.

2. Locating a root by change of sign

If f is continuous on $[a, b]$ and $f(a)$ and $f(b)$ have **opposite signs**, then $f(x) = 0$ has at least one root in (a, b) (spec ref e1).

Worked example. Show that $x^3 - 2x - 5 = 0$ has a root between 2 and 3.

$$f(2) = 8 - 4 - 5 = -1, \quad f(3) = 27 - 6 - 5 = 16$$

f changes sign and is continuous on $[2, 3]$, so a root lies in that interval.

State **all three** parts: both function values, the fact that the sign changes, and that f is continuous. The continuity clause is the one usually omitted, and it is what makes the argument valid.

Showing a root to a given accuracy

Worked example. Show that a root is 1.86 correct to 2 decimal places.

Test the **bounds of the rounding interval**: evaluate $f(1.855)$ and $f(1.865)$. A sign change between them means the root lies in $[1.855, 1.865)$, and every value there rounds to 1.86.

When change of sign fails

Situation	What goes wrong	Example
Repeated root — the curve touches the axis	No sign change despite a root	$f(x) = (x - 2)^2$
Vertical asymptote in the interval	Sign change but no root	$f(x) = \frac{1}{x - 2}$, with $f(1) = -1$, $f(3) = 1$
Several roots in the interval	Signs at the ends may agree, hiding them all	Two roots between the endpoints

The asymptote case is the subtle one: $f(1)$ and $f(3)$ genuinely have opposite signs for $\frac{1}{x-2}$, yet there is no root at all. The discontinuity, not a root, causes the change — which is exactly why continuity must be stated.

3. Fixed point iteration

Rearrange $f(x) = 0$ into the form $x = g(x)$, then iterate $x_{n+1} = g(x_n)$ from a starting value (e3).

Worked example. Rearrange $x^3 - x - 4 = 0$ for iteration.

One rearrangement: $x^3 = x + 4$, so $x = \sqrt[3]{x + 4}$, giving

$$x_{n+1} = \sqrt[3]{x_n + 4}$$

The **same** equation can be rearranged in several ways, and they do not all behave alike — one may converge while another diverges from the same starting value. Convergence depends on the rearrangement, not only on the equation.

Convergence and the diagrams

Near a root, the iteration converges when $|g'(x)| < 1$ and diverges when $|g'(x)| > 1$. The behaviour has a standard picture drawn between $y = g(x)$ and $y = x$:

Diagram	Occurs when	Appearance
Staircase	$g'(x) > 0$	Steps approach the root from one side
Cobweb	$g'(x) < 0$	Spirals inwards, alternating either side of the root

Finding the limit

Worked example. To what does $x_{n+1} = \sqrt{x_n + 2}$ converge?

At the limit L , both x_{n+1} and x_n equal L :

$$L = \sqrt{L + 2} \implies L^2 - L - 2 = 0 \implies (L - 2)(L + 1) = 0$$

The terms are positive, so $L = 2$.

4. The Newton-Raphson method

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Geometrically, follow the **tangent** at the current point down to where it meets the x -axis, and take that as the next estimate (e4).

Worked example. Apply Newton-Raphson to $f(x) = x^3 - 5$ from $x_0 = 2$.

$$f(2) = 3, \quad f'(x) = 3x^2 \Rightarrow f'(2) = 12$$

$$x_1 = 2 - \frac{3}{12} = 1.75$$

A second iteration: $f(1.75) = 0.359375$ and $f'(1.75) = 9.1875$, so

$$x_2 = 1.75 - \frac{0.359375}{9.1875} = 1.7109 \text{ (4 d.p.)}$$

Compare $\sqrt[3]{5} = 1.7100$ — two steps already give three-figure accuracy.

Newton-Raphson converges much faster than a simple iteration: near a simple root the error is roughly **squared** at each step, so the number of correct digits approximately doubles.

When Newton-Raphson fails

- $f'(x_n) = 0$ — the tangent is horizontal and never meets the axis; the method breaks down entirely.
- **Starting near a stationary point** — the tangent is nearly horizontal, so the next estimate is thrown far away, possibly converging to a different root or diverging.
- **Poor starting value** — convergence is only guaranteed sufficiently close to the root.

Both failures are best explained with a **sketch** showing the tangent. A description that says "it fails" without showing why the tangent misbehaves rarely earns the marks.

5. The trapezium rule

For n strips of width $h = \frac{b-a}{n}$ with ordinates $y_0, y_1, \dots, y_n$:

$$\int_a^b y \, dx \approx \frac{h}{2} \left[y_0 + y_n + 2(y_1 + y_2 + \dots + y_{n-1}) \right]$$

Note that n strips need $n + 1$ ordinates. The first and last are counted once; every interior one is doubled.

Worked example. With 4 strips over the interval from 1 to 3, the strip width is

$$h = \frac{3 - 1}{4} = 0.5$$

requiring ordinates at $x = 1, 1.5, 2, 2.5, 3$.

Over- or under-estimate

Each trapezium is formed by a **chord** joining two points on the curve. Whether that chord lies above or below the curve settles the direction of the error (spec ref c34).

Curve on the interval	Chords lie	Trapezium rule gives
Concave upwards	Above the curve	An over -estimate
Concave downwards	Below the curve	An under -estimate

Justify the direction by referring to the **shape** — "the curve is concave upwards on this interval, so each chord lies above it and the estimate is too large". Simply asserting "overestimate" with no reason scores nothing.

Halving the strip width reduces the error by a factor of roughly 4.

Rectangle bounds

Sums of rectangles give upper and lower **bounds** on the area (c35). For a monotonic function, rectangles drawn using left-hand and right-hand ordinates bracket the true value between them.

6. Common pitfalls

- Omitting the continuity condition when arguing from a change of sign.
- Testing the rounded value itself rather than the bounds of the rounding interval.
- Confusing the number of strips with the number of ordinates.
- Doubling the first and last ordinates in the trapezium rule.

- Asserting over- or under-estimate without referring to the concavity.
- Writing the Newton-Raphson formula with the fraction inverted.
- Working in the wrong calculator mode when f involves trigonometric functions.
- Assuming any rearrangement $x = g(x)$ will converge.